CDIS – Y25

Department of CSE

B.Tech. CSE Trimester -5

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Cloud Infrastructure and Services

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LAB RECORD

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Experiment -1 (b)

Enabling Multi-Factor Authentication to Secure Your Access

To enable Multi-Factor Authentication (MFA) for an AWS account to provide an additional layer of security during the login process.

Description

Multi-Factor Authentication (MFA) adds an additional verification step to the AWS sign-in process. After entering the IAM user credentials, the user verifies their identity using a registered passkey/security key or an authenticator application. This experiment creates an IAM user, signs in as that user, configures MFA, and tests the MFA-protected login.

Prerequisites

• An active AWS account.

• Valid AWS account login credentials.

• Access to a smartphone for authentication (if using a virtual MFA application).

Steps

Part A: Creating an IAM User

1. Open a web browser and navigate to the AWS Management Console.
2. Sign in to the AWS account using the root user credentials.
3. In the AWS Management Console search bar, type IAM and select Identity and Access Management (IAM).
4. From the left navigation pane, select Users.
5. Click Create user.
6. Enter a username (for example, StudentUser01).
7. If the console provides an option to enable console access, select Provide user access to the AWS Management Console.

8. Set a custom password or allow AWS to generate a password (Autogenerated password). 9. Select the

option requiring the user to create a new password at the next sign-in, if appropriate. Click Next. 10. Under

Set permissions, select Attach policies directly.

11. Select AdministratorAccess (AWS managed job function policy). Click Next.
12. Review the username and permission details. Tags are optional.
13. Click Create user.
14. Note down the IAM user sign-in details and securely save the generated password, if applicable. You can download the .csv file.
15. Click Return to user to view the created IAM user.
16. Sign out of the root account.

Part B: Signing in as an IAM User

1. Open the AWS Management Console sign-in page and select IAM user as the sign-in type. Alternatively, use the Console sign-in URL saved in the downloaded CSV file.
2. Enter the AWS account ID or account alias. This may be filled automatically when using the Console sign-in URL.
3. Enter the newly created IAM username and password from the downloaded CSV file.
4. If prompted, change the temporary password and set a new password.
5. Verify that you can successfully access the AWS Management Console.

Part C: Enabling Multi-Factor Authentication (MFA)

1. Sign in to the AWS Management Console using the IAM user credentials.
2. Search for IAM in the AWS Management Console search bar and open the IAM dashboard.
3. From the left navigation pane, select Users.
4. Click the newly created IAM username.
5. Open the Security credentials tab.
6. Locate the Multi-Factor Authentication (MFA) section.
7. Click Assign MFA device.
8. Enter a device name (for example, StudentMFA01).
9. Select the MFA device type: Passkey or Security key, or Authenticator app (Google Authenticator or Microsoft Authenticator).

Option A: Passkey or Security Key

1. Select Passkey or Security Key and click Next.
2. Follow the browser instructions to create and register the passkey.
3. If prompted, select a password manager such as Google Password Manager to save the passkey. 4.

Complete verification using the available method (biometric authentication, device PIN, or security key). 5.

Click Continue to complete registration.

Option B: Authenticator App

1. Select Authenticator app and click Next.
2. Open an authenticator application on your smartphone (Google Authenticator or Microsoft Authenticator).
3. Select the option to add a new account.
4. Scan the QR code displayed in the AWS console.
5. The authenticator application generates a six-digit authentication code.
6. Enter the first six-digit MFA code displayed in the application.
7. Wait for the application to generate the next code.
8. Enter the second six-digit MFA code.
9. Click Add MFA or Assign MFA to complete setup.
10. Verify that the selected MFA device is displayed as assigned to the IAM user.

Part D: Testing MFA Authentication

1. Sign out of the AWS Management Console and open the AWS sign-in page.
2. Select IAM user sign-in.
3. Enter the AWS account ID or account alias.
4. Enter the IAM username and password.
5. Complete MFA verification using the configured device:
6. If Passkey or Security Key was selected, choose sign-in using the registered passkey and complete verification using the registered device, biometric authentication, or PIN.
7. If Authenticator App was selected, open the authenticator application and enter the current six-digit MFA

authentication code.

8. Verify that you can successfully log in to the AWS Management Console after completing MFA verification.

Output

• An IAM user is successfully created in AWS.

• Appropriate permissions are assigned to the IAM user.

• Multi-Factor Authentication is successfully enabled for the IAM user.

• The IAM user can securely access the AWS Management Console using the configured MFA method.

Result: Multi-Factor Authentication was configured for the IAM user, and the login process was tested using the selected MFA method.

